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Three bearing positioning system

Abstract

A three-bearing system (TBS) running on a hexagonal rail behind a false wall enables a TV hung on a first wall to move horizontally along the first wall, traverse an inside corner formed by the first wall and a second wall, and move horizontally along the second wall. A yoke supports the three bearings and is attached to the TV by a rigid support arm (made of metal, carbon fiber, or other structural support material) that has passed through one or more horizontal slots in a false wall. The protruding support arms holding the TV in place are attached to the hexagonal rail system that is permanently attached to the structure of the building. Through the user's selected movement or positioning, viewing the TV is easier and more comfortable from many positions in a room.

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Background/Summary

TECHNICAL FIELD

[0001] This invention relates to positioning systems. More particularly, this invention relates to positioning a television (TV) or other flat panel visual display device on one or more flat surfaces.

BACKGROUND

[0002] Flat panel devices such as large screen TVs, computer displays, pictures, paintings, and the like normally are fixedly mounted to a wall surface, a ceiling, or a fixed horizontal surface such as a credenza top, table top, bureau top, or a wall niche. If one desires to change the location of the flat panel display device on a wall, one has to remove the flat panel display device from its mounting on the wall, remount it in the new location, and then repair the damage to the wall where the visual display device originally was mounted. If one desires to put the TV back to where it was, or to a new location, the process has to be repeated. The task of safely hanging a large screen TV requires a minimum of three people, two people to lift the TV and another person to direct the two people holding the large screen TV to reposition the TV for best TV viewing. It would be advantageous to be able to reposition a large screen TV without having to go to such lengths.

[0003] Published and Allowed US Patent Application US 2021/0354383 A1 refers, in FIG. **14**, to a biaxial positioning system for moving a flat panel object hung from a flat surface in two dimensions. FIG. **15** of the Published Application refers to a uniaxial embodiment of the positioning system. The apparatus in the published patent application cannot (1) traverse an inside corner, and (2) does not provide a mechanism to support straight or curved false walls. False walls provide permanent cover for the mechanical components so that these components cannot to be seen from the room side of the false wall. The published patent application does not address how a hidden linear rails can guide a TV straight or around an inside curve. The published patent application does not address hiding rigid support components (made of metal, carbon fiber or other structural support material) required to move the TV behind “floating” false walls. The published patent application has the TV mounted to an exposed multi-linear, non-parallel, and non-coplanar rail system, which is mounted to a room's structural support wall, and not a hidden hex rail system mounted to a room's structural support wall.

SUMMARY

[0004] Applicant has solved the problems of the prior positioning systems by devising a positioning system whereby a flat panel display device can be slid horizontally across a flat surface without removing it from its mounting. In addition to allowing movement with respect to the flat surface on which it is mounted, Applicant's invention allows movement of the flat panel display device to an adjacent flat surface around an inside corner formed by the flat surface on which the visual display device originally was mounted and an adjacent flat surface on which the visual display device is to be newly located. The mechanism to accomplish this can be hidden behind a false wall to maintain the aesthetics of the room environment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** is a front view of an example of a positioning system in accordance with this invention.

[0006] FIG. **1**A is another front view of the apparatus of FIG. **1** illustrating what's behind the TV and the false wall shown in FIG. **1**.

[0007] FIG. **2** is a side sectional view of the example of the positioning system of FIG. **1** taken along section line **2-2** shown in FIG. **1**.

[0008] FIG. **3** is a top sectional view of the example of the positioning system of FIG. **1** taken along section line **3-3** shown in FIG. **1**.

[0009] FIGS. **4***a*, **4***b*, and **4***c* are top views of the example of the positioning system of FIG. **1**

illustrating traversal of an inside corner formed by two intersecting adjacent walls.
[0010] FIG. **5** depicts a side view of a magnified yoke assembly.
[0011] FIG. **6** is a perspective view of a yoke.
[0012] FIG. **7**A is an exploded view of a top load bearing and two fixtures that attach the top load bearing to a yoke.
[0013] FIG. **7**B adds a yoke to the exploded view of FIG. **7**A.
[0014] FIG. **8** is a perspective view of a yoke assembly to which is attached a top load bearing and two alignment bearings.
[0015] FIG. **9**A is an exploded view of a hex rail and three bearings illustrating how the bearings ride on the hex rail.
[0016] FIG. **9**B is a perspective view of a complete yoke assembly riding on a hex rail.
DETAILED DESCRIPTION
[0017] FIG. **1** shows a rectangular flat panel visual display device **10** mounted on a vertically disposed wall. The flat panel display device **10** can be a television (TV), a computer monitor, a picture, a photo, a painting, or similar visual display device. The rest of the discussion herein will assume that the visual display device **10** is a TV, but the invention is not limited to TV's. A false wall or hung wall **12** comprising a top section **12***a*, a middle section **12***b*, and a bottom section **12***c* hides the structure of a positioning system described below. As shown most clearly in FIG. **1**A, support elements **30***a*, **30***b*, **30***c*, and **30***d* at each of the four corners of the TV **10** extend through gaps **14***a* and **14***b* in the false wall **12**. The four support elements secure the TV **10** to the wall and connect it to the positioning system behind the false wall **12** in FIG. **1**.
[0018] FIG. **2** is an edge view of the apparatus of FIG. **1** taken from the right hand side of FIG. **1**. FIG. **2** shows a vertical wall **15** commonly used in conventional building construction. The vertical wall **15** comprises a series of vertical 2×4 or 2×6 wooden studs **16** covered with one or more layers of gypsum board or sheetrock **18**. In this example of the invention, the sheetrock **18** is covered with a layer of plywood **20** that provides a secure anchor surface for the positioning system shown in FIG. **2**.
[0019] Standoff box frames **22***a*, **22***b*, and **22***c* are attached to the plywood layer **20**. They connect the plywood layer **20** to the backside of the false wall **12** and provide support for the wall **12**. In addition, box frames **22***a* and **22***b* support hex rails **24***a* and **24***b*. The size of the box frame standoffs **22***a*, **22***b*, and **22***c* is such that an appropriate amount of space is provided between the false wall **12** and the plywood layer **20** to accommodate the positioning equipment shown in FIGS. **2** and **3** described below. The box frames **22***a*, **22***b*, and **22***c* can be made of any suitable structural material such as aluminum.
[0020] The positioning system comprises two hex rails **24***a* and **24***b* that extend horizontally along wall **15** behind the false wall **12**. The two hex rails **24***a* and **24***b* extend parallel to the gaps **14***a* and **14***b*. The hex rails **24***a* and **24***b* are located on adjacent walls **15** and **17** shown most clearly in FIGS. **4***a*, **4***b*, **4***c*, and **4***d*. As also shown in FIGS. **4***a*, **4***b*, **4***c*, and **4***d*, the hex rails **24***a* and **24***b* each comprise a first straight portion **25***a* that extends horizontally across the wall **15**, a curved portion **29***a* that extends around an inside corner **19**, and a second straight portion **27***a* that extends horizontally across another wall **17** that intersects wall **15**. Wall **17** is similar to wall **15**. Like wall **15**, wall **17** supports a false wall that has upper and lower gaps as in FIG. **1**. Although not shown in FIGS. **4***a*, **4***b*, and **4***c*, there is another rail **24***b* underneath rail **24***a*. Rail **24***b* is configured like rail **24***a*.
[0021] Four identical yoke assemblies **26***a*, **26***b*, **26***c*, and **26***d*, one for each of the four corners of the rectangular TV **10**, support the TV **10** and connect it to the rails **24***a* and **24***b* as shown in FIG. **1**A. Yoke assembly **26***a* connects the upper right hand corner of the TV **10** in FIG. **1** to the upper rail **24***a*. Yoke assembly **26***b* connects the lower right hand corner of the TV **10** in FIG. **1** to the lower rail **24***b*. Yoke assembly **26***c* connects the upper left hand corner of the TV **10** in FIG. **1** to the upper rail **24***a*. Yoke assembly **26***a* connects the lower left hand corner of the TV **10** in FIG. **1** to

the lower rail **24***b*. The yoke assemblies ride along the rails **24***a* and **24***b* and carry the TV **10** horizontally with them. Rail **24***a* comprises a first straight portion **25***a* that extends horizontally along wall **15**, a second straight portion **27***a* that extends horizontally along wall **17**, and a curved portion **29***a* connecting the first and second portions of rail **24***a*. Rail **24***b* is underneath rail **24***a* in FIGS. **4***a*, **4***b*, and **4***c* and comprises a first straight portion **25***b* that extends horizontally along wall **15**, a second straight portion **27***b* that extends horizontally along wall **17**, and a curved portion **29***b* connecting the first and second portions of rail **24***c*. As shown in FIGS. **4***a*, **4***b*, and **4***c*, the TV **10** can be positioned horizontally, either manually or by means of a motor drive, along straight portions **25***a* and **25***b* of the rails **24***a* and **24***b* and across wall **15**. Similarly the TV **10** can be positioned horizontally along straight portions **27***a* and **27***b* of rails **24***a* and **24***b* and across another wall **17** that intersects wall **15**. The TV **10** can be moved horizontally from wall **15** to **17** and from wall **17** to wall **15** across an inside corner **19** formed by the intersection of adjacent walls **15** and **17**. FIG. **4***a* shows the TV **10** traveling horizontally along wall **15** and approaching the inside corner formed by walls **15** and **17**. FIG. **4***b* shows the TV **10** traversing the inside corner. FIG. **4**C shows the TV **10** exiting the inside corner and beginning its horizontal movement along wall **17**.

[0022] FIG. **5** is a magnified view of one of the four yoke assemblies **26***a*, **26***b*, **26***c*, or **26***d* referred to above that illustrates how the yoke assemblies are connected to the hex rails **24***a* and **24***b*. The other three yoke assemblies are the same. Each yoke assembly comprises an inverted U-shaped yoke **28***a*, **28***b*, **28***c*, or **28***d* more clearly shown in the perspective of FIG. **6**. Support elements **30***a*, **30***b*, **30***c*, and **30***d*, shown most clearly in FIG. **1**A, are attached to the top of a respective yoke **28***a*, **28***b*, **28***c*, or **28***d*. The support elements **30***a*, **30***b*, **30***c*, and **30***d* extend through the false wall **12** through the slots **14***a* and **14***b* and are connected to the corners of the TV **10**.

[0023] FIG. **5** shows that the hex rails **24***a* or **24***b* each comprise a rectangular base **32**, a stem portion **34** protruding from the top of the base portion **32**, and a hexagonal cross-section top portion **36** sitting atop the stem portion **34**. The hexagonal top portion **36** provides three bearing surfaces that interact with a tri-bearing system (TBS) mounted on the yoke **28***a*, **28***b*, **28***c*, or **28***d*. The bearing surfaces are each angularly separated from the other two bearing surfaces by an angle of 120 degrees. Each of the three bearings in the TBS defines axes of rotation separated from one another by the same 120 degrees.

[0024] A bearing **38** is located between the arms of the inverted U-shaped yoke **28***a*, **28***b*, **28***c*, or **28***d*. Bearing **38** rolls along the top surface **39** of the hexagonal top of the rail **24***a* or **24***b*. The top surface thus provides a load-bearing surface for the positioning system of this invention. Bearings **40** and **42** located at the ends of the arms of the inverted U-shaped yoke roll along surfaces **41** and **43**, respectively, and maintain proper alignment of the yoke assembly as it moves along the hex rails **24***a* and **24***b*.

[0025] The bearings **38**, **40**, and **42** are standard roller element bearings, for example, ball bearings or roller bearings. These bearings comprise two different diameter concentric rings that are rotatable with respect to one another. The concentric rings thus form a bearing race. Balls or rollers are confined between the rings that facilitate smooth rotation of the bearing. The outer circumference of each of the bearings in FIG. **5** may be covered with a layer of elastomeric or rubber-like material **44**, **46**, and **48** that prevents slippage with respect to rails **24***a* and **24***b*. The bearings **38**, **40**, and **42** thus roll along their respective bearing surfaces **39**, **41**, and **43** similar to the way automobile tires roll along a road. This layer of elastomeric or rubber-like material on ball bearings **38**, **40**, and **42** offers quieter, smoother, rumble free movement, and adds a greater lifespan of the bearing surfaces on the hex rails **24***a* and **24***b*.

[0026] FIG. **6** is a perspective view of one of the yokes **28***a* in the yoke assemblies. Yokes **28***b*, **28***c*, and **28***d* are the same as yoke **28***a* in FIG. **6**. Yoke **28***a* is an inverted U-shaped structure comprising vertical arms **52** and **54** connected by a top horizontal portion **56**. The ends of the arms **52** and **54** end in outwardly flared portions **58** and **60**. Vertical arms **52** and **54** contain cylindrical openings **62** and **64** for the receipt of two ends of a roller element bearing **38** that supports the TV

10. The flared portions **58** and **60** have cylindrical openings **66** and **68** for the receipt of the alignment bearings **40** and **42**. The support element **30***a* is attached to the top portion **56** of the yoke **28***a* in FIG. **6**.

[0027] FIG. **7**A is an exploded view of the top bearing **38** and two fixtures **70** and **72** that attach the top bearing **38** to one of the yokes **28***a*. FIG. **7**B adds the yoke **28***a* to the parts shown in FIG. **7**A. Attachment of the other top bearings to their respective yokes **28***b*, **28***c*, and **28***d* is similar to that shown in FIGS. **7**A and **7**B. Bearing fixtures **70** and **72** extend through openings **62** and **64** in the yoke **28***a* to provide a cylindrical shaft **74** extending across the arms **52** and **54** of the yoke **28***a*. When fully assembled, the cylindrical shaft **74** extends through a circular opening **75** in bearing **38**. The bearing **38** spins about this shaft **74** when the yoke **28***a* is moved along hex rail **24***a*. Top bearings associated with yokes **28***b*, **28***c*, and **28***d* are structured and operate in the same way that bearing **38** associated with yoke **28***a* is structured and operates.

[0028] FIG. **8** shows the top bearing **38** supported between the arms **52** and **54** of the yoke **28***a*. Two alignment bearings **40** and **42** bolted to flared portions **58** and **60** of the yoke **28***a*. FIG. **9** shows the yoke assembly shown in FIG. **8** attached to the hex rail **24***a*. The top bearing **38** rolls on the bearing surface **39**, the alignment bearing **40** rolls on the bearing surface **41**, and the alignment bearing **42** rolls on bearing surface **43**. The other three yokes **28***b*, **28***c*, and **28***d* have three bearings attached in the same way as bearings attached to yoke **28***a*. Those other three yokes **28***b*, **28***c*, and **28***d* are attached to a respective hex rail **24***a* or **24***b*, as the case may be, in the same way that the yoke **28***a* is attached to hex rail **24***a*.

Highlights of This Positioning System

[0029] To recap, the intended purpose of the Tri-Bearing-System (TBS) described above is to enable a TV **10** or other flat panel visual display device to move horizontally along two walls **15** and **17** and traverse an inside corner formed by the two intersecting walls **15** and **17**. Yoke assemblies **26***a*, **26***b*, **26***c*, and **26***d* are attached to the four corners of the rectangular TV **10** by rigid support arms or swivels **30***a*, **30***b*, **30***c*, and **30***d*. The support arms **30***a*, **30***b*, **30***c*, and **30***d* are rigid strips of support material. The support elements may be made of metal, carbon fiber, or other suitable structural support material able to support the weight of the TV **10**. Each of the support elements **30***a*, **30***b*, **30***c*, and **30***d* is rotatable about the end points at which the support arm is attached to a respective yoke **28***a*, **28***b*, **28***c*, and **28***d* in a respective yoke assemblies **26***a*, **26***b*, **26***c*, and **26***d*. The support elements **30***a*, **30***b*, **30***c*, and **30***d* are also rotatable about the end point connected to the TV **10**. The support arms thus act as swivels that permit the TV to traverse an inside corner formed by two intersecting walls **15** and **17**. Each support arm passes through one of a pair of horizontal slots **14***a* and **14***b* in a false wall **12** that hides the mechanism of this invention. The protruding support arms holding the TV **10** in place are attached to a hexagon (hex) rail system, which in turn is permanently attached to a structural wall of the building. A user can easily and selectively move or position a TV **10** for more comfortable viewing from many different positions in a room. Because this positioning system is (1) a permanent structure, (2) is secured to the building by screws and adhesives, and (3) adds or enhances the value of the property, it qualifies as a “Capital Improvement” for tax purposes.

Inside Corner Movement

[0030] The positioning system can traverse an inside corner or a straight-line wall due to the constant positioning, repositioning, and interaction of a main top load supporting roller element bearing, and the two lower guide or alignment roller element bearings. The main top bearing carries the compression load exerted by the TV **10** and its components. Two lower alignment or guide bearings keep the top main load bearing centered on the top surface of the hex shaped track or rail. The two lower guide bearings act as horizontal movement limiters or stabilizers, by pressing against the two opposite surfaces of the straight or curved hex guide rail.

Straight-Line Movement

[0031] A top load supporting roller element bearing **38** and two alignment roller element bearings

40 and **42** working in tandem with one another along the hex rails **24***a* and **24***b* enable straight-line movement of the TV **10**. The two alignment bearings **40** and **42** constantly maintain the top bearing's position on the hex rail, by limiting the top bearing's movement either right or left from the centerline of its respective hex rail **24***a* or **24***b*.

Hidden Mechanical Components Behind a False Wall

[0032] The mechanical parts of the positioning system for the wall mounted TV **10** are attached to the hex rail fixture **24***a* and **24***b*. The hex rail fixture lies behind the false wall **12** and is mounted to the building's structure. A supporting element at each corner of the TV **10** is attached to the hex rail fixture and protrudes through the false wall **12** to the room side of the false wall **12**. The distance between the parallel horizontal slots **14***a* and **14***b* is wholly dependent on the size of the TV that is to be mounted on the wall. Two horizontal parallel slots **14***a* and **14***b* enable all support equipment in the positioning system to pass through the false wall **12** into the room side for attachment to the TV **10**.

[0033] The purpose of carbon box frames shown in the drawings is to support the false sheetrock wall **12** in a way that does not impede the movement of the TV **10** and positions the false sheetrock wall **12** so that adequate space between the false wall **12** and the existing wall **15** or **17** accommodates the positioning system. The number of box frames used depends on how many are needed to support the hex rails **24***a* and **24***b* and also how many are needed to provide adequate rigidity to the false wall **12**.

[0034] All of the mechanical mechanisms, which move with the TV **10** on the upper and lower hex rails **24***a* and **24***b*, are hidden behind a false sheetrock wall **12**. TV support elements **30***a*, **30***b*, **30***c*, and **30***d* pass through the false sheetrock wall to the TV. Two horizontal slots **14***a* and **14***b*, each of which may approximately 0.5-inches wide, are formed in the false sheetrock wall **12**. A first of these horizontal slots is located below the top carbon fiber box frame, and the second horizontal slot is located above the bottom carbon fiber box frame. One of the slots is slightly above the top TV swivels, and the other slot is slightly below the bottom TV swivels. The horizontal slots in the false wall enable the four arms connected to and projecting outward from hex rail yoke assemblies to be fastened to the swivels mounted on the right-and left-handed sides of the top and bottom of the TV.

[0035] The overall visual effect is, the TV can move horizontally while being suspended from the four protruding yoke extension arms, without being fastened to the wall. The effect is created by having the mid-section of the false wall secured independently to the carbon fiber standoffs from behind the wall . . . that is, without using the buildings 2×4s as a fastening point.

Conclusion

[0036] The Title, Technical Field, Background, Summary, Brief Description of the Drawings, Detailed Description, and Abstract are meant to illustrate the preferred embodiments of the invention and are not in any way intended to limit the scope of the invention. The scope of the invention is solely defined and limited in the claims set forth below.

Claims

1. A positioning system adapted to support a planar object on first and second adjacent planar surfaces, the first and second planar surfaces being oriented at a predetermined angle with respect to one another so as to define an inside corner, the positioning system comprising: an elongated rail comprising a first linear portion extending in a first direction along the first planar surface, a second linear portion extending in a second direction along the second planar surface, and a curved portion connecting the first and second linear portions; and a support structure adapted to connect the planar object to the elongated rail.

2. The positioning system of claim 1, in which the elongated rail and the support structure are such that: the planar object is movable in the first direction with respect to the first planar surface; the planar object is movable in the second direction with respect to the second planar surface; and the planar object is movable from one of the planar surfaces to the other of the planar surfaces around the inside corner defined by the first and second adjacent planar surfaces.

3. The positioning system of claim 1, in which: the elongated rail has a hexagonal cross section that defines three bearing surfaces; the support structure comprising a yoke that has first, second, and third roller element bearings, each of which respectively engage first, second, and third bearing surfaces defined by the hexagonal cross section of the elongated rail.

4. The positioning system of claim 3, in which each of the first, second, and third roller element bearings define a respective axis of rotation, each of the axes of rotation being at an angle of approximately 120 degrees with respect to the other two axes of rotation.

5. The positioning system of claim 1, in which the planar object is a display device.

6. The positioning system of claim 1, in which the planar object is a television.

7. The positioning system of claim 1, in which the planar object is a picture.

8. The positioning system of claim 1, in which the first and second planar surfaces are room surfaces oriented at a right angle with respect to one another.

9. A positioning system, adapted to support a planar object on adjacent first and second planar surfaces, the first and second planar surfaces being oriented at a predetermined angle with respect to one another so as to define an inside corner, the positioning system comprising: a first elongated rail, suspended from the first and second planar surfaces, comprising a first linear portion extending in a first direction along the first planar surface, a second linear portion extending in a second direction along the second planar surface, and a first curved portion connecting the first and second linear portions; a second elongated rail, suspended from the first and second planar surfaces, the second elongated rail being oriented parallel to the first elongated rail, the second elongated rail comprising a third linear portion extending in the first direction along the first planar surface, a fourth linear portion extending in the second direction along the second planar surface, and a second curved portion connecting the third and fourth linear portions; and a support structure adapted to connect the planar object to the first and second elongated rails.

10. The positioning system of claim 9, in which the first and second elongated rails and the support structure are such that: the planar object is movable in the first direction with respect to the first planar surface; the planar object is movable in the second direction with respect to the second planar surface; and the planar object is movable from one of the planar surfaces to the other of the planar surfaces around the inside corner defined by the adjacent first and second planar surfaces.

11. The positioning system of claim 9, in which: the first elongated rail has a first hexagonal cross section that defines a first set of three bearing surfaces; the second elongated rail has a second hexagonal cross section that defines a second set of three bearing surfaces; and the support structure comprises: a first yoke having first, second, and third roller element bearings, each of first, second, and third roller element bearings respectively engaging first, second, and third bearing surfaces defined by the hexagonal cross section of the first elongated rail; and a second yoke having fourth, fifth, and sixth roller element bearings, each of fourth, fifth, and sixth roller element bearings respectively engaging fourth, fifth, and sixth bearing surfaces defined by the hexagonal cross section of the second elongated rail; and in which each of the first, second, and third roller element bearings define a respective first, second, and third axis of rotation, each of the first, second, and third axes of rotation being at an angle of approximately 120 degrees with respect to the other two of the first, second, and third axes of rotation; and each of the fourth, fifth, and sixth roller element bearings define a respective fourth, fifth, and sixth axis of rotation, each of the fourth, fifth, and sixth axes of rotation being at an angle of approximately 120 degrees with respect to the other two of the fourth, fifth, and sixth axes of rotation.

12. The positioning system of claim 11, further comprising: a first support element adapted to connect the first yoke to a top edge of a rectangular planar display device; and a second support element adapted to connect the second yoke to a bottom edge of a rectangular planar display device.

13. The positioning system of claim 11, in which the support structure further comprises: a third yoke having seventh, eighth, and ninth roller element bearings, each of seventh, eighth, and ninth roller element bearings respectively engaging the first set of three bearing surfaces defined by the hexagonal cross section of the first elongated rail and a fourth yoke having tenth, eleventh, and twelfth roller element bearings, each of the tenth, eleventh, and twelfth roller element bearings respectively engaging fourth, fifth, and sixth surfaces in the second set of three bearing surfaces defined by hexagonal cross section of the second elongated rail.

14. The positioning system of claim 13, in which: each of the seventh, eighth, and ninth roller element bearings define a respective seventh, eighth, and ninth axis of rotation, each of the seventh, eighth, and ninth axes of rotation being at an angle of approximately 120 degrees with respect to the other two of the seventh, eighth, and ninth axes of rotation; and each of the tenth, eleventh, and twelfth roller bearings define a respective tenth, eleventh, and twelfth axis of rotation, each of the tenth, eleventh, and twelfth axes of rotation being at an angle of approximately 120 degrees with respect to the other two of the tenth, eleventh, and twelfth axes of rotation.

15. A positioning system adapted to permit positioning a visual display device suspended on one of more flat surfaces horizontally across the one or more flat surfaces, comprising: a hex rail, having a hexagonal cross section that defines at least three bearing surfaces, the hex rail having a first straight portion that extends horizontally across a first flat surface, a second straight portion that extends horizontally across a second flat surface, and a curved portion connecting the first straight portion to the second straight portion, the first flat surface and the second flat surface forming an inside corner; and a tri-bearing yoke assembly connected to the hex rail comprising a yoke supporting three roller element bearings, each of the three roller element bearings traveling along one of the three bearing surfaces.

16. The positioning system of claim 15, further comprising: a false wall covering the hex rail and the tri-bearing yoke assembly.